Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 9 · SLO 2 — Solved Worksheet

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Unit â€“ 2 : Session â€“ 9 : SLO â€“ 2 | Doubly Linked List â€” Introduction

I. Fill in the Blanks

• In a doubly linked list, each node contains TWO pointers.
• The two pointers in a doubly linked list point to the PREVIOUS and the NEXT nodes.
• The pointer to the next node in a doubly linked list is commonly called NEXT (RLINK).
• The pointer to the previous node in a doubly linked list is commonly called PREV (LLINK).
• In a doubly linked list, the previous pointer of the head node is always set to NULL.
• In a doubly linked list, the next pointer of the tail node is always set to NULL.
• Insertion at the beginning of a doubly linked list requires updating the PREV pointer of the old head node.
• Deletion of a node in a doubly linked list requires adjusting the pointers of its PREVIOUS and NEXT nodes.
• Traversing a doubly linked list can be done in both FORWARD and BACKWARD directions.
• A doubly linked list allows for more efficient deletion operations compared to a SINGLY linked list.